



BottleSumo Competition

Rules & Guidelines Framework

Section 01: Competition Overview

Section 1.1: Competition Rounds

The Competition is run in Two Rounds, similar to the Global Round:

Round	Purpose
Time Trial	Robots push Bottles Off the table individually. Used to Rank Robots and seed the Tournament Bracket.
Single Elimination Round	Head-To-Head Matches, no Bottles involved.

Section 1.2: Definitions

- **Game:** A single Head-To-Head Round between Two Robots.
- **Match:** A series of Games (Best Of 3) to determine which Team Advances.
- **Match Winner:** The first Team to win Two (2) Games.
- **Robot Fall:** Occurs the moment the Robot's **Main Chassis** loses all contact with the table surface and any part of it touches the floor. A Robot still touching the table with at least one contact point, even while tipping, has **Not Fallen**.
- **Bottle Fall / Bottle Off Table:** Occurs the moment no part of the **Bottle's Base** remains in contact with the table surface, regardless of whether it has reached the floor yet.
- **Intact:** No functionally significant part (see Section 8.9) has detached from the Robot.





Section 02: Robot Requirements

Section 2.1: General Rules (All Categories)

- Robots must be **Designed, Built, and Programmed** by the **Student Team Members** themselves. Mentors/Coaches may Guide and Teach but may not build or code the Competition Robot.
- If a Robot is found to be substantially similar to another Team's Robot (including Teams from the same School), the Team is subject to Investigation (Design/Code Interview) and possible Penalty or Disqualification.
- The Robot must be fully **Constructed** and **Functional** upon arrival at the Competition.
- The Robot must be fully **Autonomous** — no Remote Control, Bluetooth/Wi-Fi Tele-Operation, or Human Signalling of any kind once the match starts.
- **One Robot Per Team**, used for the entire Competition.
- Labelling Requirements:
 1. Team ID clearly visible on any external surface (Team Name optional).
 2. A permanent "**Front**" marking on the side that carries the Primary Sensors; this side must remain the "front" for the whole Event.
- Teams must bring their own Laptop/Computer and Charging Cable to modify code between Rounds.





Section 2.2: Category Specifications

Specification	Explorer	Innovator
Maximum Robot Weight	2.5 kg	3.0 kg
Maximum Robot Size	Must Fit in a 20 x 20 x 20 cm Box	Must fit in a 30 x 30 x 30 cm Box
Expansion After Start	Allowed, as long as the robot stays within the max box size at all times.	
Number of Motors (Drive + Mechanism)	Maximum 2	Maximum 4
Number of Sensors	Maximum 3	No Limit
Battery Voltage Limit	$\leq 11.1V$ (e.g., 3S Li-ion/LiPo)	$\leq 12.6V$
Microcontroller / Brain	One Only	No Limit
Wheels / Treads / Legs	Standard Wheels or Tank Treads only. No Vacuum, Suction, or Sticky/Adhesive Material.	
Robot Shape	Front, Rear, and Side Edges Below 25mm from the ground must not be Sloped or Shaped as a Ramp. All Motors must remain Visible/Accessible for Inspection.	

Section 2.3: Sensor & Motor Requirements

- Every Robot must have at least **One Edge/Line Sensor** capable of detecting the Light/Dark Contrast of the table edge, and at least **One Object-Detection Sensor** (e.g. Ultrasonic, IR, or ToF) capable of sensing an Opponent or Bottle in front of it.
- On-Board Camera/Vision-Based Navigation (e.g., ESP32-CAM, Smartphone-Based Vision) is allowed only in the **Innovators Category**.
- Any Motor Type is allowed (DC Gear Motor, Servo, Stepper) as long as it fits within the Category's Motor Count and Voltage Limits.





Section 2.4: Materials

- Any Construction Material is allowed: Plywood, Acrylic, PVC Sheet, 3D-Printed Plastic, Cardboard-Reinforced-With-Glue, Aluminium Sheet, Etc. Tape, Glue, Screws, and Rubber Bands may be used freely.

Section 03: Playing Field Specifications

- A Standard Rectangular table, approximately **75cm × 150 – 180cm** (a single large classroom Table, Two Standard School Desks pushed together, or a 4×8 ft Plywood Sheet mounted on Sawhorses/Table Legs).
- The Table Surface will be **Light in Color** (White, Gray, Beige, or Light Brown) and reasonably Flat and Smooth. If the Natural Surface is unsuitable, it will be covered with a Light-Colored, Matte, Non-Glossy Sheet (Chart Paper, Vinyl, or Laminate) taped securely at the edges.
- The exact Color/Brightness of the surface will be confirmed on Competition Day — teams should build robots with adjustable **Sensor Thresholds** (Calibratable in code) rather than Hard-Coded values.
- The table has no **Walls, Lip, or Barrier**, all edges are open Drop-Offs.

Section 04: Bottle Specifications (Used for Time Trials)

- Standard **2-Liter Plastic Soft Drink Bottle** (e.g., Coca-Cola, RC, or similar)
- **Five (5) Bottles** are used in every Time Trial run.
- The Bottle is wrapped in **Red Paper**.
- Bottle is filled with **Sand or Fine Gravel**:
 - **Explorer Category**: 0.5 kg fill weight
 - **Innovator Category**: 1.0 kg fill weight
- The exact shade of Red Paper is not disclosed until Competition Day — Robots should not rely on an exact Color Code but on general Contrast/Reflectivity detection.
- **Arrangement**: Bottle starting positions are set by the Judge before each Round and kept Identical for every Team competing in that Round. Positions will be spaced so that no Bottle starts within **15cm** of the table edge and no two bottles start within **20cm** of each other, to keep placement Fair and Repeatable across Teams.





Section 05: Robot Start Task

- Every Robot must have a **3-Second Delay** before any part of it begins to move, for both Time Trials and Head-to-Head Games.
- All Calibration Values (Sensor Thresholds, Motor Speeds, Initial Movement Pattern) must be **Preprogrammed** before impounding ends. No On-Field Button-Press Calibration or adjustment is allowed after impounding.
- Robots must be started the same way every time (same Button/Switch) for the entire Event.
- Any preset opening motion (e.g., "turn left 90° at start" or "spin until an object is detected") must be identical for every **Game** — Teams may not change their opening strategy Round to Round.

Section 06: Competition Day Procedures

1. **Access:** Only registered Participants may access the Pit Area, Team Tables, and Practice/Competition Fields during Setup, Work Time, and Breaks.
2. **Work Period:** Immediately after the opening ceremony, a **15-Minute Timer** starts for Final adjustments.
3. **Inspection & Impound:** After the Work Period, all Robots are inspected for Weight, Size and Labelling, then impounded.
 - Judges will record the exact method used to start the Robot.
 - Battery charging is **NOT** allowed during impound.
4. Participants must not touch their Robot until instructed by a Judge.
5. **Post-Run Holding:** Immediately after a Team completes its individual Time Trial Run, the Robot is held by the Pit Judge and is not returned to the Team until the official 15-Minute Group Adjustment window (see section 7.9) begins for all Teams.

Section 07: Time Trial Round Rules

1. At the start of each Time Trial, the Judge announces the Robot's starting **Location** and **Orientation**; this stays the same for all Teams during the Event.
2. Teams place their Robots exactly as instructed.





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3. Bottle positions are announced after **Impound** and kept identical for every Team in that Round. (See Section 4)
4. The Judge measures the time taken for the Robot to push all **5 Bottles** off the table, timed to **1/100th** of a second, starting from zero.
5. **Maximum Time Allowed: 2 Minutes (120 Seconds).**
6. If the Robot falls off the table or fails to push off all bottles, the Survival Time and Number of Bottles pushed off are recorded instead.
7. **Stability Requirement & Penalty:** The Robot must remain intact and on the table for at least 3 Seconds after any Bottle is pushed off before that push is considered final.
 - The 3-Second Check Window itself does not count toward the recorded time.
 - If the Robot falls off the table during this 3-Second Window, the Bottle just pushed still counts toward the total Bottle pushed off.
 - A Robot that falls off the table at any point during the Round (including during a Stability Check) receives a flat **+5 Second Penalty** added to its recorded Survival Time, used only for seeding purposes (Section 7.10).
8. Timing stops at the earliest of:
 - The 5th Bottle is knocked off ("**Completion Time**")
 - The Robot falls off the table ("**Survival Time**")
 - The Robot is still on the table and running at the 2-Minute Mark ("**120-Sec Survival Time**")
9. After all **Teams** finish the Time Trial, Teams get an additional **15 Minutes** to adjust their Robot/Program before the Tournament Bracket is finalized.
10. **Seeding Order** is based on:
 - Successful completion of the Start Task
 - Number of Bottles pushed off
 - Staying on the table at least 3 Seconds after the last Bottle push
 - Completion Time or Survival Time (Lower/Higer as appropriate)





Section 08: Game (Head-to-Head) Rules – Two Robots, No Bottles

1. Maximum **2 Minutes** Per Game.
2. At the start of each Game, the Judge announces the Location and Orientation of both Robots — this may Differ from Game to Game.
3. Teams place their Robots as instructed.
4. Robots must observe the **3-Second Delay** before moving.
5. If a Robot fails to move, it automatically loses — unless both Robots fail to move, in which case it is a **Tie**.
6. If a Robot fails the 3-Second Delay requirement, it automatically loses — unless both robots fail it, in which case it is a **Tie**.
7. After the start, all Students and Judges must stay at least **1 Meter** away from the table edges until the Game ends.
8. If any part of a Robot detaches (See Section 8.9) and falls to the floor, the **Opposing Robot** is immediately declared the **Winner**.
9. **Detachment Rule:** If any **functionally significant** part of a Robot detaches and falls to the floor, the opposing Robot is immediately declared the Winner. "Functionally significant" means any Component that contributes to Locomotion, Sensing, or Game Strategy (e.g. a Wheel, Drive Track, Pushing Arm, Sensor Module, Battery Pack). Purely Cosmetic or Incidental Hardware (a Loose Zip Tie, a Decorative Panel, a Single Non-Structural Screw) does **Not** trigger this rule.
10. **One Battery Swap** is allowed Per Match if needed (e.g. due to Low Charge), at the Judge's discretion.
11. **Between-Game Repairs:** A Team may repair Mechanical Damage Sustained *during that match* between Games of the same match (e.g. reattaching a part that broke off), subject to all of the following:
 - Only existing Components may be reattached or repaired — no new Components may be added.
 - No Code changes and no Recalibration of Sensor Thresholds, Motor Speeds, or Start Behavior are permitted.
 - The Judge must visually confirm the repair before the next Game begins and log it on the Score Sheet.
 - Repair requests that would change Calibration or Code are **Denied**.





Section 09: Determining the Winner of a Game

A Robot wins a Game if any of the following occurs:

- It satisfies the Start Task and the opponent Fails it.
- It pushes the Opponent off the table and then remains intact and on the table for at least 3 Seconds.
- It remains intact and on the table for at least 3 Seconds after the opponent has fallen off, including when the opponent **Falls** or **Drives Off** the table for any reason, with no contact from this robot (**Self-Elimination**).
- The opponent has a Functionally Significant part detach per Section 8.9.
- If the Outcome is unclear, the Game is declared a **Tie** (see Section 10).

The Judge must confirm timing with a visible Stopwatch, Phone Timer, or Display Timer before declaring a Result.

Section 10: Tiebreaker rules

A Game is declared a **Tie** if:

- Both Robots touch the floor at the same moment (except the "**Detached Part**" rule in Section 8.9).
- Both Robots fall off the table within 3 Seconds of each other.
- No progress is made for 20 Seconds, at the Judge's discretion.
- Both Robots fail to move, or both fail the Start Task, or one fails to move while the other fails the Start Task.
- There is no Winner after 2 Minutes.
- The result is genuinely Unclear or too close to call.

Match-Level Tie-Breaking (if Tied after 3 Games):

1. Compare Time Trial results.
2. Play up to 2 Additional Games if Time Trial results are also Tied.
3. Final Tiebreaker: A Single-Bottle Mini Time Trial.

The Judge's decision is final in any situation Not Explicitly covered by this Rulebook. Judges may **Disqualify** any **Team** that violates rules or does not follow instructions.



Section 11: Sample Judging Rubric for Time-Trial Round

Rank	Team ID	Description	Start Task Correct?	Number of Bottles Pushed Off?	3 Sec After Last Bottle?	Completion Time (Elapsed) Robot Pushes Off All Bottles	Survival Time (Elapsed) Robot Off Table or Bottles Left	Comments
1	6	Last Bottle Off at 50s (50s Completion Time)	Y	5	Y	50		All Bottles Off, Ranked by Completion Time
2	5	Last Bottle Off at 60s (60s Completion Time)	Y	5	Y	60		All Bottles Off, Ranked by Completion Time
3	7	Last Bottle Off at 100.5s (100.5s Completion Time)	Y	5	Y	100.5		All Bottles Off, Ranked by Completion Time
4	11	Last Bottle Off at 120s, Robot Falls Off at 122s	Y	5	N		122	All Bottles Off, But Did Not Survive For 3 Sec, Ranked by Survival Time
5	2	Last Bottle Off at 118s Robot Falls Off Table at 119s	Y	5	N		119	All Bottles Off, But Did Not Survive For 3 Sec, Ranked by Survival Time



Section 12: FAQ (Frequently Asked Questions)

Q: Can our Robot have multiple programs to choose from at the start of a game?

A: No. The same Program and Start Method must be used for every Game.

Q: Robot A pushes Robot B off, but A also falls off before 3 seconds. Who wins?

A: Tie Game.

Q: Robot A fails the Start Task, Robot B succeeds and survives 3 seconds. Who wins?

A: Robot B Wins.

Q: Can the Explorers use block-based coding (like Scratch/mBlock) instead of C++?

A: Yes. Any Programming Language or Environment is allowed in every Division.

Q: Is it okay if our Robot expands after the game starts?

A: Yes, as long as it never exceeds the Maximum Box Size for its Category at any point.

Q: Are pneumatics or flywheel mechanisms allowed?

A: Yes, in all Categories, as long as they meet Size, Weight, Motor Count, and Safety Rules.

Q: Will bottle color/table surface be revealed before the event?

A: The exact Shade/Brightness is confirmed only on Competition Day, but general Table Dimensions and Surface Type will be published in advance so Teams can Calibrate.

Section 13: Contact Us

For any Queries Regarding the Competition, Please Contact:

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